

Multiple Choice

1. **Answer:** C

Options: A) $O(1)$; B) $O(n)$; C) $O(n^2)$; D) $O(\log n)$

Note: $O(n^2)$ is the largest asymptotic notation among the choices because it represents a quadratic growth rate, which increases faster than linear ($O(n)$) or logarithmic ($O(\log n)$) growth rates as the input size n becomes very large.

2. **Answer:** A

Options: A) ['Hi!', 'Hi!', 'Hi!', 'Hi!']; B) ['Hi!'] * 4; C) Error; D) None of the above.

Note: The correct answer is ['Hi!', 'Hi!', 'Hi!', 'Hi!'] because the expression ['Hi!'] * 4 in Python creates a new list containing four copies of the original list element.

3. **Answer:** B

Options: A) sort('ab'); B) sorted('ab'); C) "ab".sort(); D) 1/0

Note: The expression 'sorted('ab')' is valid in Python 3.5 because it uses the built-in 'sorted()' function to return a list of the characters in the string 'ab' in ascending order.

4. **Answer:** C

Options: A) $O(\log e^N)$; B) $O(\log N)$; C) $O(\log \log N)$; D) $O(N)$

Note: $O(\log \log N)$ represents a growth rate that increases very slowly compared to other common functions such as polynomial or exponential functions, making it the slowest-growing type of function among the options provided.

5. **Answer:** D

Options: A) 1; B) 2; C) 3; D) 4

Note: The correct answer is 4 because the max() function in Python returns the largest element from the list, and in the list [1, 2, 3, 4], the largest number is 4.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: In Python 3, the function 'cmp()' does not exist; instead, the correct function to determine the count of items in a list is 'len()'.

7. **Answer:** True

Options: A) True; B) False

Note: An Internet Protocol (IP) address is assigned to a device to uniquely identify it on a network, enabling communication and data exchange when it connects to the Internet.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the expression `"a[i] == max || !(max != a[i])"` is logically equivalent to `"a[i] == max"` rather than simplifying to `"a[i] != max."`

9. **Answer:** True

Options: A) True; B) False

Note: The `strip()` function in Python 3 removes all leading and trailing whitespace characters from a string by default, which confirms that the statement is true.

10. **Answer:** False

Options: A) True; B) False

Note: Negative indexing in Python 3 allows access to characters in a string by counting from the end, so it does not always result in an error.

Long Form Response

11. **Answer:** `def tn_ap(a,n,d): | return tn`

Note: The function correctly calculates the t-nth term of an arithmetic progression by utilizing the formula $tn = a + (n - 1) * d$, where 'a' is the first term, 'n' is the term number, and 'd' is the common difference.

12. **Answer:** `def adjac(ele, sub = []): | def get_coordinates(test_tup): | return (res)`

Note: The answer correctly defines a function that utilizes a helper function to generate and return all coordinates that are adjacent to a specified coordinate tuple, demonstrating an understanding of coordinate manipulation in programming.

End of Answer Key